Printed Pages:01 **Sub Code: RCA-501**

Paper Id: 231533 **Roll No.** | | | | | | | | | | |

MCA
(SEM V) THEORY EXAMINATION 2022-23
COMPUTER GRAPHICS & ANIMATION

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

(a) Point out various display methods of computer graphics.
(b) What do you mean by scan conversion of line?
(c) Define the terms window and viewport.
(d) What is 4 bit code algorithm?
(e) What do you mean by hidden surface removal?
(f) Define geometric continuity of curves.
(g) What do you mean by computer generated animation?

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

(a) How the line can be drawn using Bresenham's algorithm? Discuss with the help of an example.
(b) By taken an example discuss various types of 2D reflections.
(c) Discuss various types of parallel projections.
(d) Write short note on Bezier curves.
(e) Discuss story boarding method of animation.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) With the help of an example, discuss Bresenham's circle drawing algorithm.
(b) Explain flood filling algorithm.

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Discuss Sutherland Hodgeman polygon clipping algorithm.
(b) Whether 2D translation follow commutative property or not? Discuss.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Discuss Scan line algorithm of hidden surface removal.
(b) Differentiate between image space and object space methods.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) What are the various shading? Discuss any one of them.
(b) Write short note on filtering of image.

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write short note on procedural methods of animation.
(b) Discuss various motion control methods of animation.

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